Protection Follows the Label: How Large Language Model Résumé Screeners Turn the Motherhood Penalty into a Caregiver Penalty

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Abstract

Large language models (LLMs) now screen job applicants, and their developers train them not to treat people differently by gender. We ask what such systems do with the signals that make up the motherhood penalty. On a résumé, motherhood bundles a protected category (being a woman) with signals that are correlated with gender but not protected as such: an interruption in employment and visible family responsibilities. We propose that in aligned models, *protection follows the label*: the more directly a signal names a protected category, the more carefully models handle it, while correlated signals are penalized freely. Study 1, an audit of four widely used models (616,419 decisions, 28 occupations), unbundles the signal on otherwise identical résumés and places each component along this gradient. Signals of caregiving carry a large penalty, for mothers and fathers alike, visible only when applicants are compared: a parent returning from leave wins a head-to-head choice against an equally qualified childless applicant 11 to 31 percent of the time. The penalty is unchanged when all gender information is removed. Explicitly named gender elicits small, vendor-specific behavior—correction toward women, parity, or refusal—rather than a penalty. An instruction that names caregiving gaps reduces the gap penalty but not the penalty on family signals it leaves unnamed. Study 2, planned, tests whether the gradient emerges with alignment by comparing the same open-weight model before and after post-training. Because women carry caregiving signals more often than men, a screener that protects the label can reproduce the motherhood penalty in effect.

1 Introduction

Mothers pay a price in the labor market that fathers do not. They earn less than comparable childless women (Budig and England 2001), and in audit and laboratory studies employers rate them as less competent and committed, recommend them for hire less often, and offer them lower salaries, while fathers are not penalized (Correll, Benard, and Paik 2007). The penalty persists when mothers present direct evidence of commitment, which suggests that it rests on norms about what mothers should do as well as on inferences about what they can do (Benard and Correll 2010).

Hiring is now partly delegated to large language models, and these are not neutral copies of the texts they learned from. Developers train them, through instruction tuning and preference optimization, to avoid treating people differently by protected characteristics such as gender, and evaluate them on whether outputs change when a protected attribute changes. This raises a question that audits of human employers could not pose: what happens to a form of discrimination when the decision-maker has been trained against its category but not against its correlates?

The motherhood penalty is a natural case, because on a résumé motherhood is a bundle. It combines a protected category—the applicant is a woman—with signals that are correlated with gender but not protected as such: time out of work, and visible family responsibilities such as children, parenting activities, or a caregiving leave. In human hiring these components travel together, and audits cannot remove gender while keeping the family situation visible. An LLM screener can be shown the same résumé thousands of times with one component added or removed, including with every trace of gender removed, so the bundle can be taken apart.

We propose that in aligned models *protection follows the label*. Signals differ in how directly they refer to a protected category (Figure 1): an explicit statement of gender names it; a gendered first name implies it; a caregiving gap or a line about children is merely correlated with it. If alignment teaches models to handle named categories carefully, behavior should differ along this gradient: careful, developer-specific treatment where gender is named, weaker effects where it is only implied, and unchecked penalties where the signal is a correlate that no one has trained against. The same logic predicts how remedies work: an instruction should protect what it names and leave untouched what it does not.

Study 1 tests these expectations in an audit of four widely used models. It unbundles the motherhood signal on identical résumés, varies how directly gender is conveyed, observes the result at the decision margins where selection happens, and tests the fairness instructions that deployers commonly add. The findings line up with the gradient. Caregiving signals carry a large penalty that falls on fathers as much as on mothers, is unchanged when gender information is removed, and appears only when applicants are compared. Explicitly named gender elicits small and vendor-specific behavior—correction toward women in one model, parity in another, refusal in a third—rather than a penalty. An instruction that names caregiving gaps reduces the gap penalty but not the penalty on family signals. Study 2, which we describe as a planned design, asks where the gradient comes from by comparing the same open-weight model before and after post-training.

The argument contributes to three conversations. For the sociology of gender and work, it separates components of the motherhood penalty that human hiring confounds and shows that, in these systems, the ideal-worker components survive while the gendered one does not (Acker 1990; Williams 2000). For research on discrimination and its regulation, it identifies a mechanism—protection that attaches to labels rather than consequences—by which discrimination can persist through correlated signals once a category is policed, a pattern with parallels in studies of employers who lose access to a signal (Agan and Starr 2018). For algorithmic auditing, it shows that audits organized around protected categories, single-applicant scores, or answered questions only would certify these systems as fair.

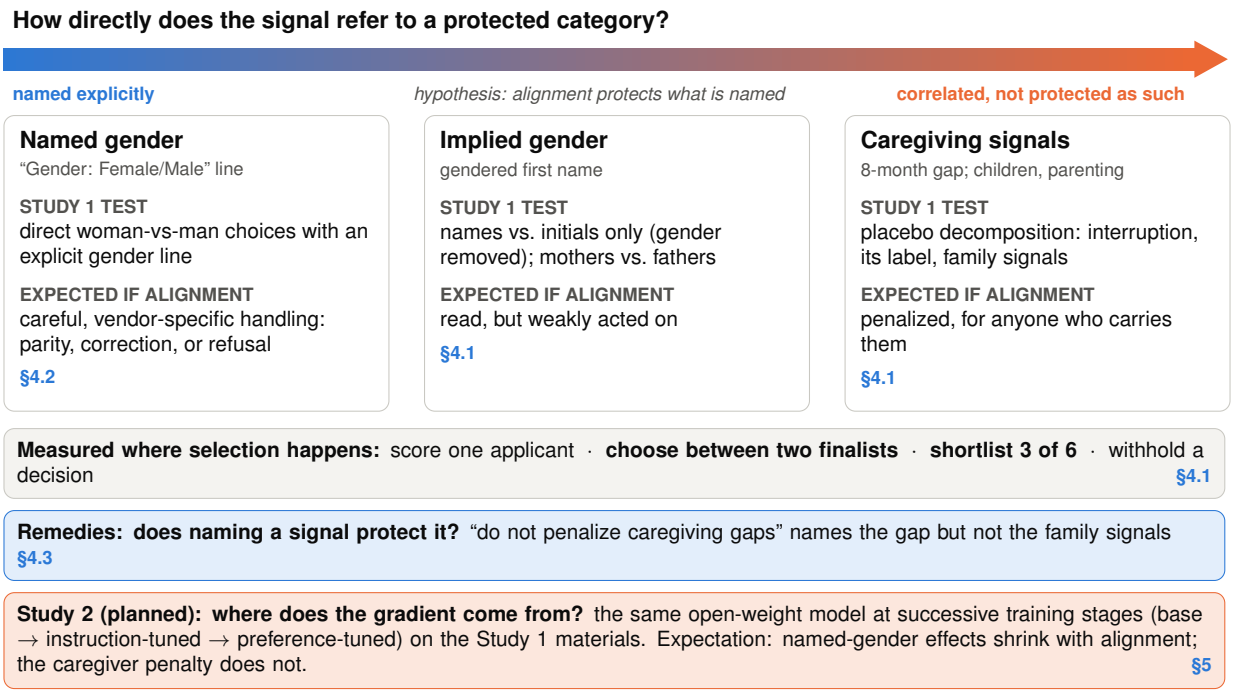


Figure 1: Framework: protection follows the label. Components of the motherhood signal are ordered by how directly they refer to a protected category, from named gender to correlated caregiving signals. Each column states the Study 1 test and what the label account expects; bands show where outcomes are measured, how remedies are tested, and the planned Study 2. Section numbers locate each test.

2 Background

2.1 The motherhood bundle

Explanations of the motherhood penalty can be read as claims about which part of the bundle does the work. *Gender-statistical accounts* hold that employers treat motherhood as a proxy for lower availability and that the inference is gendered: mothers, not fathers, are expected to reduce their commitment (Phelps 1972; Correll, Benard, and Paik 2007), and a normative variant holds that mothers violate expectations of devotion to family (Benard and Correll 2010). Both predict a penalty specific to mothers, which requires the evaluator to know the applicant's gender. *Ideal-worker accounts* locate the penalty in the organization of work around an abstract worker who is fully available and free of outside obligations (Acker 1990; Williams 2000; Blair-Loy 2003). On this view the penalty attaches to departures from continuous employment and to signs of competing obligations, whoever shows them. Men who signal caregiving are penalized too (Rudman and Mescher 2013; Williams, Blair-Loy, and Berdahl 2013), and employment gaps are penalized whatever their reason (Kroft, Lange, and Notowidigdo 2013; Weisshaar 2018). Because women take most caregiving leaves (Goldin 2014), such a penalty is gendered in its consequences even when it is not gendered in its application.

2.2 Alignment and the label

Commercial LLMs are trained after pretraining to follow instructions and to produce outputs that raters and written principles prefer, including outputs that do not treat people differently by protected attributes. Audits of automated hiring have likewise asked mostly whether outcomes differ by protected attributes signaled through names (An et al. 2025; Armstrong et al. 2024; Rozado 2025; Wilson and Caliskan 2024), following the logic of anti-discrimination law, which asks whether outcomes differ across protected categories (Barocas and Selbst 2016). Both practices are organized around the category. Neither directly targets signals that are correlated with a category but not named by it. If models learn from this kind of training to recognize and handle protected categories when they are named, their behavior toward a signal should depend on how directly it names one. Training against a category could produce caution—equal treatment, correction in favor of the protected group, or refusal—where the category is explicit, while leaving judgments about correlated signals as pretraining left them.

2.3 Expectations along the gradient

Figure 1 arranges the components of the motherhood signal by how directly they refer to gender and states what the label account expects at each position.

- *Named gender* (an explicit "Gender: Female/Male" line): careful, developer-specific handling—parity, correction toward women, or refusal—rather than a penalty.
- *Implied gender* (a gendered first name): read, but acted on weakly; removing it should change little.
- *Correlated signals* (an employment gap; children and parenting activities): penalized, for anyone who carries them, as the ideal-worker account predicts.
- *Remedies*: an instruction protects the signal it names and not the ones it leaves unnamed.

The gender-statistical account makes contrary predictions: a penalty for mothers but not fathers, which shrinks when gender is removed. The framework also says where to look. Selection is comparative, and a penalty that reflects relative judgments about availability may surface only when applicants compete, so the expectations are tested at several decision margins. We state these as orienting expectations; the tests specified before data collection are listed in Section 3.4.

3 Study 1: Design

Study 1 is a correspondence-style audit (Bertrand and Mullainathan 2004) of LLM screeners. Every estimate compares a résumé with itself, with and without a signal, so applicant quality is held constant by construction.

3.1 Unbundling the correlated signals

Table 1 lists the signals. Every application includes a short cover note, so conditions differ only in what the note and résumé say. The *parent* signal adds "married with two children" to the note and a parent-teacher association line to the résumé; *parent with leave* adds an eight-month "Parental leave" entry. Three placebos

Table 1: Unbundling the motherhood signal

Signal	What it adds to the same résumé	What the comparison isolates
None	neutral cover note	baseline
Personal (placebo)	a length-matched hobby and volunteer line	disclosure as such
Travel gap (placebo)	8-month "Career break — extended travel"	the interruption (vs. equal treatment)
Parental leave only (placebo)	the same dates labeled "Parental leave"	the caregiving label (vs. travel gap)
Parent	"married with two children"; parent-association line	parenthood without a gap (vs. personal)
Parent + leave	Parent plus the parental-leave entry	family signals on top of a leave (vs. leave only)
Neutral gap labels	same dates as "company closure" or "certification"	whether travel is stigmatized
Parent + commitment	Parent plus objective commitment bullets	the normative account
Pregnant (women only)	"currently pregnant"	pregnancy, incl. withholding

Note: Signals render identically for women and men; every application includes a cover note. Gender information is varied separately (names; name or initial plus an explicit gender line; initials only).

separate the components: a *travel gap* with the same dates isolates the interruption; *parental leave only*, without children, isolates the caregiving label; and a length-matched *personal* line isolates disclosure. The comparisons yield four components: the interruption (travel gap against equal treatment), its caregiving label (parental leave only minus travel gap), family signals on top of a labeled leave (parent with leave minus parental leave only), and parenthood without a gap net of disclosure (parent minus personal). Because travel might itself be stigmatized, a later phase substitutes two neutral labels with the same dates ("Job search after company closure," "Full-time certification program"). *Parent with commitment evidence* and a *pregnancy* signal, for women only, complete the set. The signals are deliberately explicit; the parental-leave gap is the most realistic and anchors the decomposition.

3.2 Moving along the gradient: how gender is conveyed

Gender information is varied at four levels that correspond to positions in Figure 1: the first name alone (implied); the name plus a header line stating "Gender: Female/Male" (named); a gender-neutral initial plus that line (named, without the name); and initials only (gender removed). Names are six per gender from Bertrand and Mullainathan (2004), all coded as White to hold race constant (Gaddis 2017), rotated within every condition. Direct choices between a woman and a man with the same family status measure gender preference as such. A perception check confirms that models read implied gender: whenever a model used a gendered pronoun in an explanation, it matched the name or header in every case, and gendered pronouns all but disappeared under initials only. Asked directly, however, models often answered "unknown" for names (Luna 34 percent, Haiku 58 percent) while stating explicit headers correctly in every case—an early sign that what models *say* about gender is itself a trained behavior.

3.3 Decision margins and remedies

Signals are evaluated at four margins: *scoring* one applicant (fit on a ten-point scale, interview recommendation, salary, four ratings); *choosing* between two finalists of the same gender, one with the signal and one without (counterbalanced; equal treatment = 50 percent); *shortlisting* three of six applicants, one with the signal (impact ratio relative to the same résumé without it, comparable to the four-fifths rule); and *declining to decide*—refusing or withholding a salary—recorded as an outcome. A remedies phase repeats the pairwise tasks under three system prompts: none, a generic equal-employment-opportunity prompt, and that prompt plus "do not penalize employment gaps taken for caregiving," which names the gap but not the family signals.

3.4 Models, materials, and inference

We audit four widely available, low-cost models—DeepSeek V4.1 Flash, Claude Haiku 5.5, GPT-6 Luna, and Qwen 3.8 Flash—pinned to dated versions and single endpoints, with reasoning disabled, structured output, and no system prompt except in the remedies phase. The materials are 154 résumés validated as qualified for 28 occupations, 20 spanning occupational female share (4.7 to 97.1 percent) and wages and 8 chosen to weaken the correlation between female share and care work. Data were collected in four phases between October 14 and 30, 2026 (616,419 logged calls).

Standard errors are clustered at the design's units: pairwise choices by the unordered pair of résumés (40 pairs in the main occupations, 16 in the added ones), scores by résumé, shortlists by occupation with résumé and position fixed effects. Effective sample sizes are set by résumé pairs, not calls. Tests use the Holm correction within families; claims of no difference are tested for equivalence at ± 5 percentage points; few-cluster tests are checked with wild cluster bootstrap; and main tests are re-estimated as linear probability models with controls. Responses produced by repair passes are excluded from choice analyses, and refusals and unusable answers are analyzed as outcomes and bounded (Section 4.4). Designs and confirmatory tests were specified in analysis plans committed before each paid run; the ± 5-point bound for the second phase was chosen after its data were collected but before the tests were run, and was specified in advance for the remedies phase. The study was not registered externally; other analyses are labeled exploratory, and cross-model comparisons are descriptive.

4 Study 1: Results

We begin at the correlated end of the gradient, where the label account expects the largest effects, then move to named gender and to remedies. Differences are in percentage points ("points").

4.1 Correlated signals: a caregiver penalty

It appears where selection happens. Family signals barely move the scores models give a single applicant: relative to the same résumé without the signal, the parent signals change women's fit ratings by at most 0.35 points on a ten-point scale and interview recommendations by at most 1.5 percentage points, and Haiku rates mothers 0.34 points *higher*. When applicants compete, the same signals are costly (Table 2). A mother

Table 2: Probability that the applicant with the family signal is chosen in pairwise choice (main run)

Signal — applicant	DeepSeek	Haiku	Luna	Qwen
Parent — mother	.357	.233	.297	.387
Parent — father	.349	.276	.294	.398
Parent + commitment — mother	.419	.266	.228	.407
Parent + leave — mother	.310	.110	.186	.230
Parent + leave — father	.308	.130	.184	.256
Ranking: shortlist rate relative to the same résumé without the signal				
Mothers	.71	.25	.78	.75
Fathers	.70	.23	.81	.80

Note: Pairwise choice against an equally qualified childless candidate of the same gender; .50 = equal treatment. Standard errors clustered by résumé pair (40 pairs) give 95% confidence intervals of ±.02–.07; no interval includes .50. Ranking ratios are résumé fixed-effect estimates from pools of six; .80 is the four-fifths threshold.

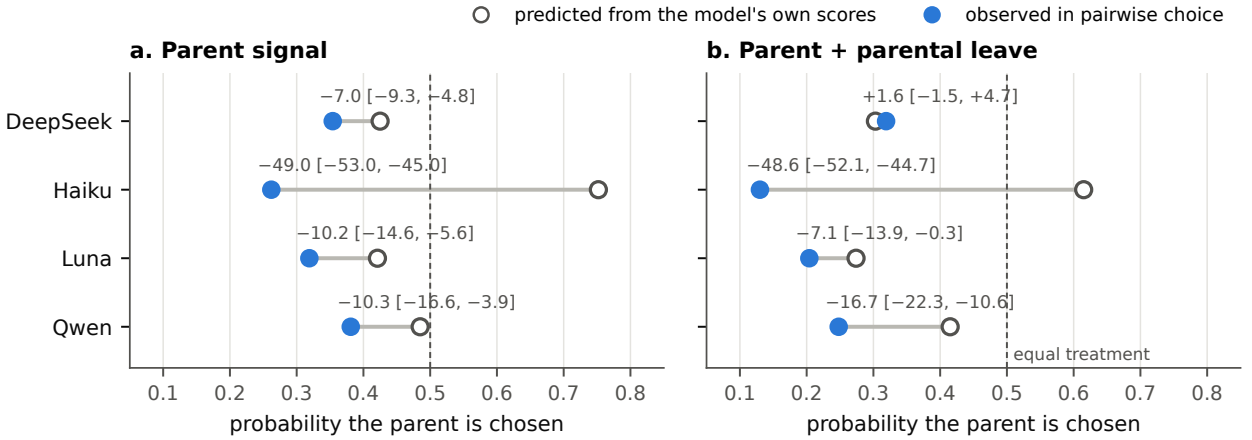


Figure 2: The caregiver penalty appears where selection happens. Hollow markers: probability that a mother is chosen over an equally qualified childless woman, predicted from the model's own scores; filled markers: observed.

returning from parental leave is chosen against an equally qualified childless woman 31 percent of the time by DeepSeek, 23 by Qwen, 19 by Luna, and 11 by Haiku—the model that scores her higher. In shortlisting, mothers reach the shortlist at 71, 75, and 78 percent of their childless rate with DeepSeek, Qwen, and Luna, each below the four-fifths threshold, and at 25 percent with Haiku. Choices are not magnified scores: a noise model built from each model's own rating variability predicts far smaller penalties than observed in seven of eight model-signal combinations (Figure 2), and the penalty remains 9 to 32 points among résumé pairs with clear quality differences.

It consists of the interruption and the family signals, not the label. Table 3 and Figure 3 decompose the parent-with-leave penalty (mothers, explicit gender, 40 résumé pairs). An eight-month gap labeled travel costs 10.5 to 26.0 points. Calling it parental leave adds nothing—the label is equivalent to zero for Haiku

Table 3: Placebo decomposition of the parent-with-leave penalty (percentage points)

Component (comparison)	DeepSeek	Haiku	Luna	Qwen
Interruption	−10.5	−18.4	−26.0	−22.4
(travel gap vs. .50)	($t = -8.0$)	($t = -8.5$)	($t = -7.3$)	($t = -8.9$)
Caregiving label	+1.7	−1.1	+6.5	+6.7
(parental leave only − travel gap)	[+0.5, +2.9]	[−3.2, +0.9]	[+3.7, +9.2]	[+4.9, +8.5]
Family signals on top of leave	−10.6	−17.4	−11.5	−12.1
(parent + leave − parental leave only)	[−12.7, −8.6]	[−20.2, −14.5]	[−15.4, −7.5]	[−14.7, −9.5]
Parenthood net of disclosure	−10.4	−19.2	−12.1	−7.3
(parent − personal placebo)	[−12.8, −8.0]	[−23.4, −15.0]	[−16.3, −7.9]	[−10.3, −4.3]

Note: Mothers; 20 main occupations; gender stated explicitly (levels L1 and L2 pooled); 40 résumé pairs. Brackets: 95% confidence intervals clustered by résumé pair. The personal placebo itself costs 2.9–3.5 points for DeepSeek, Luna, and Qwen and raises Haiku's choice share by 3.7 points.

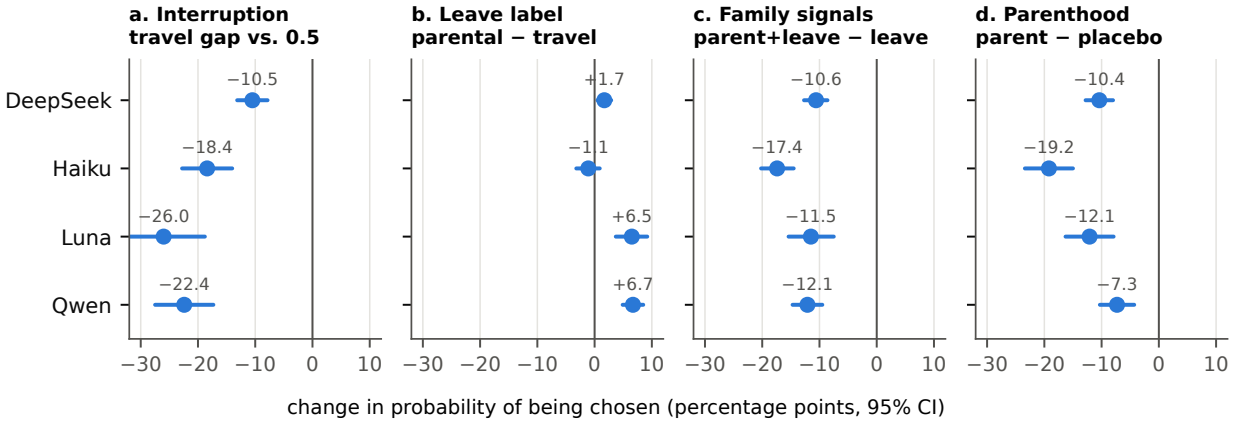
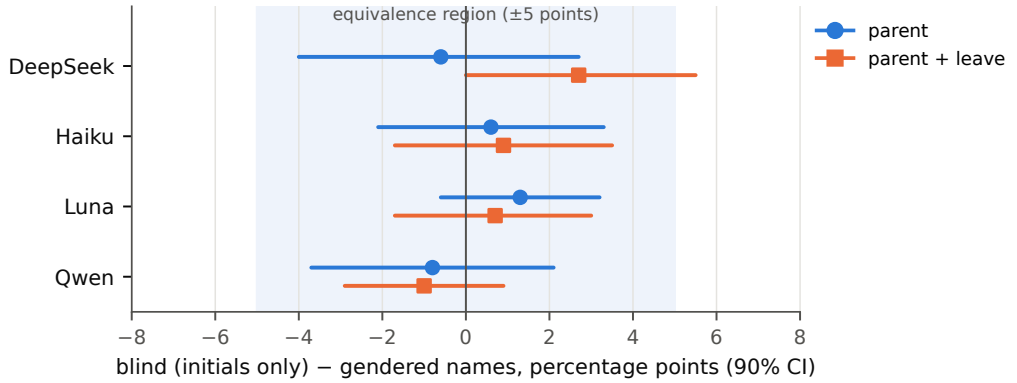


Figure 3: Decomposition of the parent-with-leave penalty: the interruption and family signals carry it; the caregiving label adds nothing.

and 1.7 to 6.7 points *less* costly than travel for the other models—and travel was not unusually stigmatized: no neutral label was significantly more costly than travel after correction, Qwen penalized travel 4.3 to 5.4 points more than either neutral label, and the largest estimate in the other direction was DeepSeek's certification label for women (4.0 points, Holm p = .14). Adding children and parenting activity to a résumé that already carries a labeled leave costs a further 10.6 to 17.4 points, and parenthood without a gap, net of the personal placebo, costs 7.3 to 19.2 points. Commitment evidence does not restore equal treatment (Table 2). The family-signal and parenthood contrasts survive Holm correction in 23 of 24 tests across six alternative samples, and "greedy" availability language in job ads deepens the leave penalty for some model–gender combinations, as an availability norm would imply (Section 4.4).

It does not depend on gender. Moving from implied gender to no gender at all changes nothing. In the added occupations, the parent penalty with gendered names differs from the penalty under initials only by +0.6 (DeepSeek), −0.5 (Haiku), −1.3 (Luna), and +0.3 points (Qwen); in seven of eight model-signal combinations the difference is statistically equivalent to zero within ±5 points, and in the eighth (DeepSeek's



Change in the parent's probability of being chosen when all gender information is removed (8 added occupations, 16 résumé pairs). A CI inside the band means the difference is statistically equivalent to zero.

Figure 4: Removing all gender information leaves the penalty unchanged.

Table 4: Mother–father differences for the parent signal (main run)

Outcome	DeepSeek	Haiku	Luna	Qwen
Pairwise choice (points)	+0.6	**−4.4**	−0.4	+0.1
	($t = 0.6$)	($t = -3.2$)	($t = -0.4$)	($t = 0.0$)
Paired salary ($)	+138	−150	−109	−16
Commitment rating, paired	+.02	**−.14**	+.00	+.12
Commitment rating, single applicant	**−.09**	**−.06**	+.03	+.00

Note: Mother minus father on the same résumé pairs (choice, paired ratings) or résumés (single-applicant ratings). Bold: significant after Holm correction within outcome.

leave signal) the interval reaches 5.6 (Figure 4). Shortlisting ratios likewise barely move across the four levels of gender information. Fathers are chosen below equal treatment in every cell of every model—a father returning from leave is chosen 31, 26, 18, and 13 percent of the time by DeepSeek, Qwen, Luna, and Haiku—and there is no fatherhood bonus. Mother–father differences are small and not significant after correction in three models (Table 4); Haiku chooses mothers 4.4 points less often than fathers (Holm p = .02).

The models' explanations follow the decomposition (exploratory). In LLM-coded explanations of pairwise choices (DeepSeek V4.1 Flash coder, Qwen 3.8 Flash second coder; $\kappa = .71$ to .93 for the six codes used, lower for availability and family on non-DeepSeek explanations), the gap was cited in 24 to 91 percent of losses for the gap signals and at most 4 percent for parent signals without a gap, and family was cited for parent signals but never for the travel gap or the personal placebo; all twelve pre-stated contrasts survived Holm correction (Supplementary Figure S1). Explanations for losing parents often called family information irrelevant—for DeepSeek and Qwen much more often when the parent lost than when the parent won (53 vs. 19 and 90 vs. 42 percent). These are stated justifications, coded without human validation, not causes.

At the correlated end of the gradient, then, the models apply a caregiver penalty: a penalty on time out of work and on visible family responsibility, shared by all four vendors in sign, borne by women and men alike.

Table 5: Direct woman-versus-man choices: probability the woman is chosen

Model	Pair	Names (L0)	Name + line (L1)	Initial + line (L2)
Qwen	childless / parents	.505 / .503	.493 / .513	**.518** / .491
DeepSeek	childless / parents	.514 / **.535**	**.540** / **.540**	**.538** / **.548**
Luna	childless / parents	**.522** / **.577**	**.587** / **.647**	**.575** / **.634**
Haiku	childless / parents	**.528** / **.544**	.526 / .498	.513 / .471

Note: Both candidates share a family status; .50 = no preference; 56 résumé pairs. Bold: differs from .50 after Holm correction (six tests per model). Qwen is statistically equivalent to .50 within ±5 points in every cell.

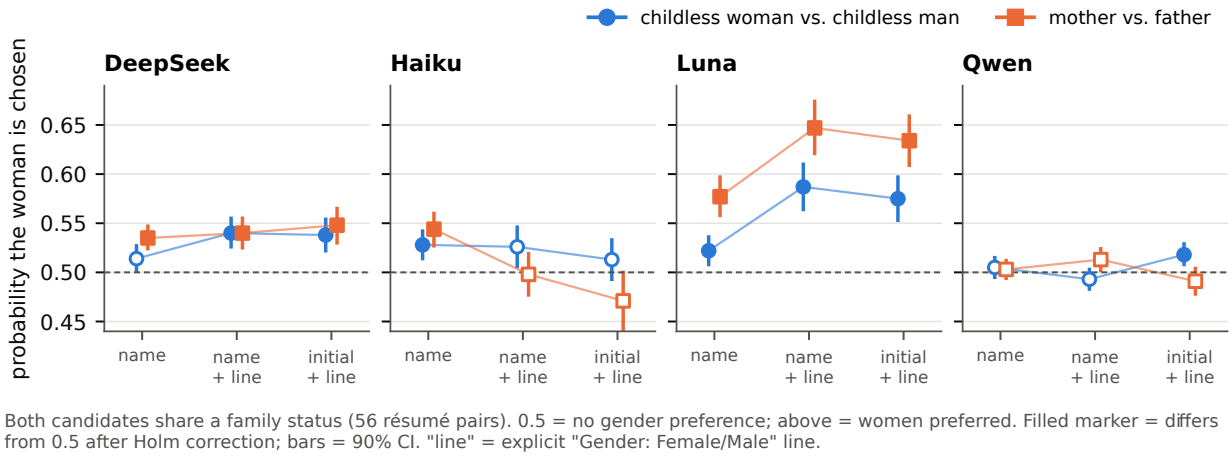
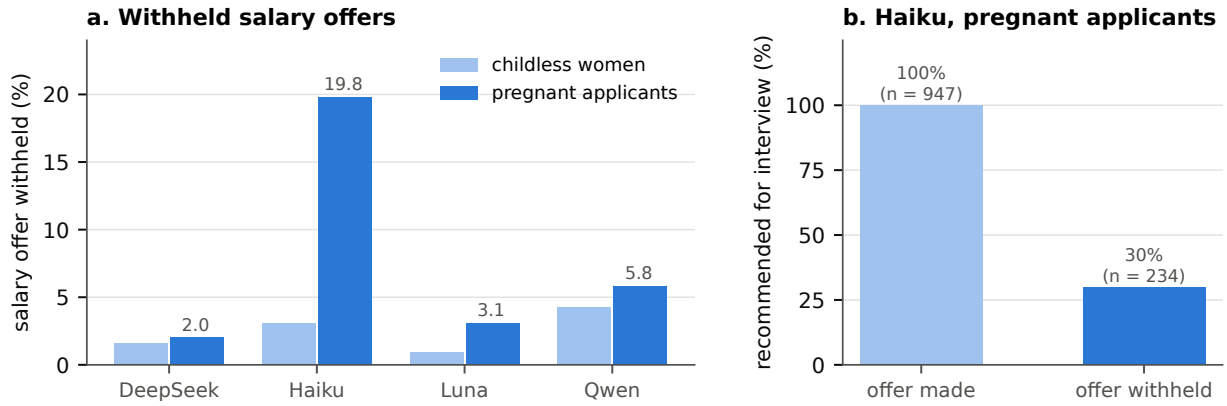


Figure 5: Named gender: direct woman-versus-man choices by how gender is conveyed.

4.2 Named gender: careful, vendor-specific handling

At the named end, the label account expects caution rather than a penalty, and behavior shaped by each developer. That is what we observe (Table 5, Figure 5). In direct choices between a woman and a man with the same family status, naming gender explicitly moves models in different directions. Luna's mild preference for women under names (.522 childless, .577 parents) grows with an explicit gender line (.587 and .647). Haiku's slight preference for mothers under names (.544) disappears with an explicit line (.498; .471 with initials). DeepSeek shows a small, stable preference for women (.535 to .548 for mothers). Qwen stays within five points of parity at every level, as equivalence tests confirm (its largest deviation, a 1.8-point tilt toward childless women under initials, is significant after correction). None of these is a penalty on women; two are corrections toward or past parity.

Caution also takes the form of declining to decide, concentrated in one vendor and in the cells where gender or family is salient. Haiku refused to choose in roughly 5 percent of mother-versus-father and parent-with-leave comparisons, against about 0.1 percent for the placebo signals. It withheld any salary for 19.8 percent of pregnant applicants (3.1 percent for childless women), and for 27.5 percent when gender was stated explicitly (Figure 6). Haiku's explanations frame withholding as protective, but it has a cost: pregnant applicants without an offer were recommended for interview 29.9 percent of the time, against every time when an offer was made, a net interview penalty of 9.4 points relative to childless women. Caution, in other



Main run, gender conveyed by name. Withholding is a refusal to propose any salary. With an explicit gender line, Haiku withholds for 27.5% of pregnant applicants.

Figure 6: Caution as declining to decide: salary offers withheld (a) and interview recommendations for Haiku's pregnant applicants by whether an offer was made (b).

Table 6: Fairness instructions (phase 4): probability that a mother is chosen, by system prompt

Model	Parent	Parent + leave	Travel gap
DeepSeek	.361 / .325 / .369	.317 / .301 / .386	.422 / .420 / .451
Haiku	.266 / .332 / .342	.074 / .195 / .217	.296 / .308 / .300
Luna	.307 / .357 / .376	.195 / .188 / .286	.238 / .283 / .307
Qwen	.381 / .333 / .351	.234 / .181 / .279	.274 / .286 / .339

Note: Entries: no system prompt / generic equal-opportunity prompt / generic prompt plus "do not penalize employment gaps taken for caregiving." Women; 20 main occupations; 40 résumé pairs; 48,000 calls; .50 = equal treatment. Hypotheses and tests were specified before the run.

words, does not guarantee equal treatment; it relocates the decision.

4.3 Remedies: an instruction protects what it names

The label account also predicts how remedies work. Table 6 reports the remedies phase (48,000 calls). A generic equal-opportunity prompt, which names gender but not caregiving, left the caregiver penalty in place in every model, with effects that depended on the vendor: none for DeepSeek, a reduction of the plain parent penalty for Luna (5.0 points), substantial reductions for Haiku (6.6 and 12.2 points), and a *deepening* for Qwen (4.7 and 5.3 points). Adding "do not penalize employment gaps taken for caregiving" reduced the interruption penalty—the signal it names—by 4 to 14 points in all four models, but left the family-signal penalty essentially unchanged except in Haiku. No arm reached equal treatment. Named gender behavior responded more readily: Luna's preference for mothers over fathers fell from .640 to .557 under the generic prompt. Naming a signal moved it toward protection; leaving it unnamed did not.

4.4 Robustness and supplementary results

Refusals and unusable answers cannot explain the caregiver penalty. Even if every missing choice had gone to the parent, the share of mothers with leave chosen in the main run could rise at most to about .32 (DeepSeek), .13 (Haiku), .22 (Luna), and .39 (Qwen); Qwen's wider bound reflects its higher rate of unusable output, which is also higher in parent cells (Round 15 formalizes these bounds). Linear probability models with controls reproduce the main tests; wild cluster bootstrap p-values change no conclusion; estimates drifted 0.8 to 2.4 points between phases, within tolerance. Supplementary results: paired salary offers are 1.7 to 2.3 percent lower for mothers; greedy availability language in job ads adds to the leave penalty for DeepSeek mothers (2.7 points) and, combined with family-friendly language, for DeepSeek applicants and Haiku mothers (5.2 to 8.0 points), while family-friendly language alone never helps; and the penalty shows no robust association with occupational female share, wages, or task composition.

5 Study 2 (planned): Where does the gradient come from?

Study 1 shows that behavior lines up with the gradient; it cannot show that alignment produced it. The four audited models are aligned commercial systems without unaligned counterparts, and their developers' training choices are not observed. Study 2 tests the mechanism directly by holding the model fixed and varying only its stage of training.

Models. Open-weight model families that release checkpoints from successive training stages—a pretrained base model, an instruction-tuned model, and a preference-tuned model—allow the same network to be compared before and after each stage (for example, the OLMo family, which publishes its post-training checkpoints and data). We will use two or more such families, at sizes that run on a single 80GB GPU, so that conclusions do not rest on one developer.

Measurement. Base models do not reliably follow instructions or produce structured output. We therefore measure choices from the models' probabilities rather than their text: for each pairwise prompt, the log-probability of answering "A" versus "B," averaged over both orderings of the candidates. The same quantity is computed for every checkpoint, so stages are compared on one scale. The Study 1 materials and conditions are reused: the placebo ladder at the correlated end, names versus initials at the implied position, and explicit gender lines and direct woman-versus-man pairs at the named end.

Predictions. If protection follows the label because of alignment, then (1) named gender effects should be present in base models and shrink, or reverse toward women, after instruction and preference tuning; (2) the caregiver penalty should be present at every stage and change little; and (3) the gap between the named and correlated ends should widen across stages. Evidence against the account would be a caregiver penalty that emerges only after alignment, or named-gender effects that are already absent in base models.

Extension. Because the models are open, Study 2 can also ask what changes inside them: whether the representation of gender that a model forms from a résumé differs across training stages, and whether caregiving signals activate that representation. Steering experiments that suppress or amplify a gender direction would test whether the caregiver penalty depends on it.

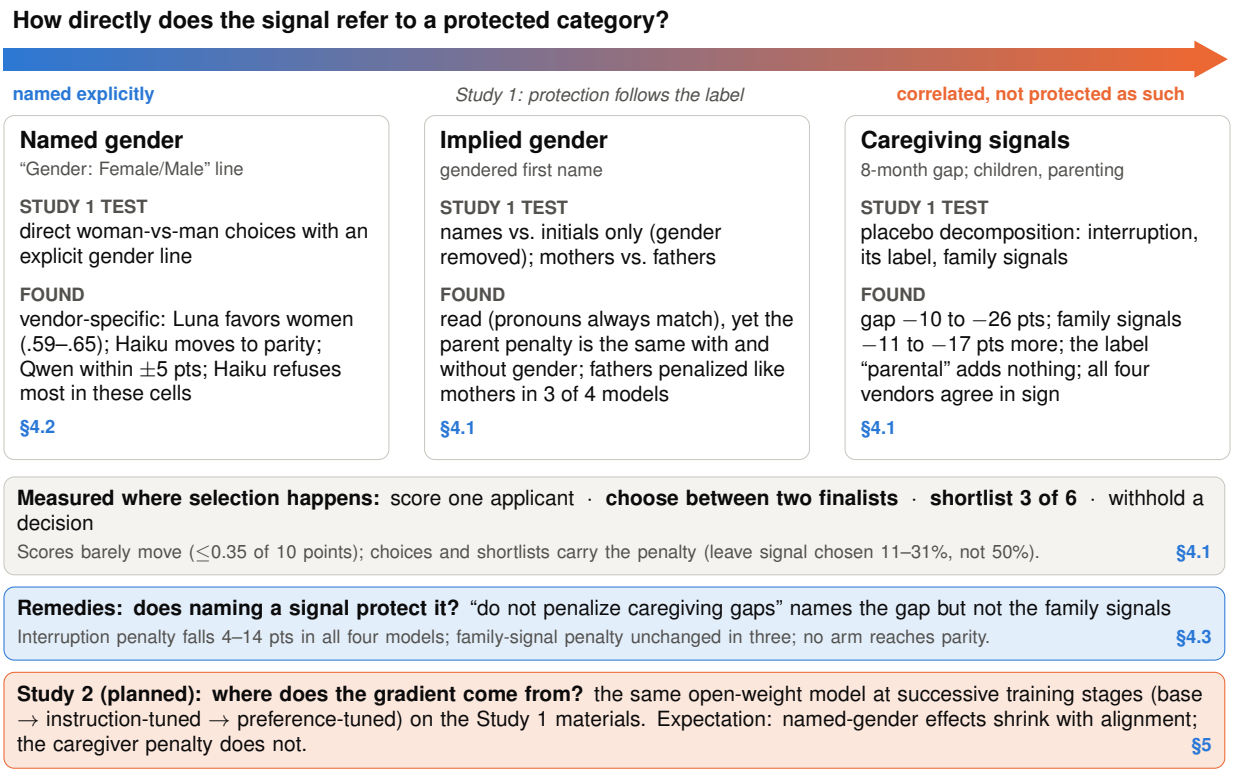


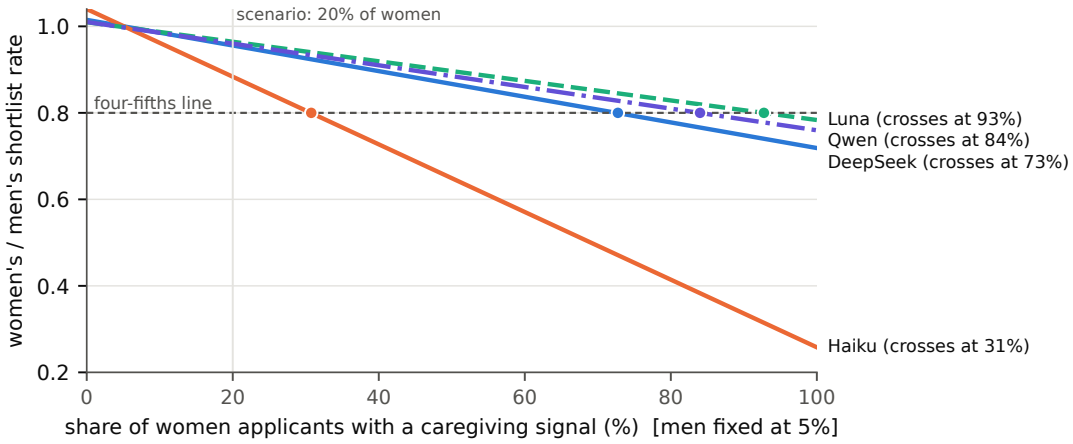
Figure 7: The framework revisited with Study 1's findings in place.

6 Discussion

6.1 The framework revisited

Figure 7 returns to the framework with Study 1's findings in place. At the correlated end, caregiving signals carry a large penalty for anyone who shows them. At the implied position, models read gender but act on it little: removing it changes nothing. At the named end, gender elicits small and vendor-specific handling—correction, parity, or refusal—rather than a penalty. And an instruction protects what it names. The gender-statistical account, which places the penalty on mothers as women, does not fit; the ideal-worker account describes the penalty that remains; and the label account describes why the remaining penalty takes the form it does.

The caregiver penalty itself has two parts with different characters. The interruption penalty behaves like an inference about availability (Phelps 1972; Kroft, Lange, and Notowidigdo 2013; Weisshaar 2018): any gap is costly whatever it is called, availability language deepens it, and naming it in an instruction reduces it. The family-signal penalty behaves like a norm: it attaches to children and parenting activity, persists on top of a gap already labeled as parental leave, is not offset by evidence of commitment, and does not respond to instructions—like the normative component of the human motherhood penalty (Benard and Correll 2010), except that it is applied to fathers too.



Applies each model's measured shortlist penalty, which is the same for women and men, to pools in which women carry the signal more often. Below 0.8 = adverse impact under the four-fifths rule.

Figure 8: Gender-neutral in form, gendered in effect: implied ratio of women's to men's shortlist rates as women's prevalence of caregiving signals rises (men fixed at 5%).

6.2 Gender-neutral in form, gendered in effect

Protection that follows the label leaves the consequences of correlated signals unprotected. Caregiving interruptions and visible family responsibilities are carried disproportionately by women (Goldin 2014), so a rule that penalizes them reproduces a gender gap through the distribution of signals rather than the treatment of individuals. Figure 8 shows how quickly this can happen: applying each model's measured shortlisting penalty to pools in which more women than men carry a caregiving signal, Haiku's gender-blind penalty produces a women-to-men shortlisting ratio below the four-fifths line once more than 31 percent of women carry the signal (men at 5 percent); the thresholds are 73, 84, and 93 percent for DeepSeek, Qwen, and Luna. Prevalence is a scenario parameter here, but the structure is the one disparate-impact doctrine was designed to capture, and the one the ideal-worker literature has long described: a norm that disadvantages women without naming them (Acker 1990; Williams 2000).

The broader implication concerns how discrimination responds to its regulation. When a protected category is policed, decisions can continue through correlated signals that are not protected as such, as research on bans of criminal-record questions has shown (Agan and Starr 2018). Alignment adds a twist: here the system itself has been trained around the category, and that training appears to make it attentive to how a signal is labeled rather than to what the signal does.

6.3 Implications for auditing

The framework explains why standard audits would miss these patterns. Audits keyed to protected categories compare women and men and find little; they need to compare applicants with and without caregiving signals. Audits that score single applicants find small effects; the penalty lives in choices and shortlists. Audits that drop non-answers miss the largest pregnancy effect, a refusal to price that carries a denial of interviews. And audits that ask models about gender measure what models are willing to say, not what they use. For

deployers, the remedies phase cautions that fairness instructions protect what they name and can backfire for some vendors.

6.4 Limitations

The family signals are explicit, and real résumés carry weaker ones. The materials are US-oriented, English, text-only, and hold race constant. Pregnancy has no male counterpart. Outcomes are recommendations, not offers or callbacks. The models are four low-cost versions at fixed dates; larger models and commercial products may differ. Pools and pairs are constructed, so the design identifies the effect of a signal, not hiring rates. The disparate-impact analysis uses scenario prevalence values. The coding of explanations relies on LLM coders without human validation. Effective sample sizes are set by résumé pairs (40 or 16 per comparison). Above all, Study 1 shows that behavior is consistent with the label account; Study 2 is needed to show that alignment produces it.

7 A Research Agenda

The label account generates questions beyond this paper. We list those we consider most informative, roughly in order of how directly they test the account.

1. *Rewording the signal.* If protection follows the label, wording that names a protected category should be treated differently from wording that does not, even when the underlying fact is the same. Candidate contrasts: "maternity leave" and "paternity leave" versus "parental leave"; "pregnancy leave" versus "medical leave"; gender-coded activities (a women's sports team, a sorority) versus neutral ones.
2. *Other protected categories along the same gradient.* Age offers a clean test: stated age (named) versus graduation year (correlated); disability: "disability leave" versus "medical leave" versus an unexplained gap; race: stated race versus names versus neighborhoods or institutions. The account predicts the same ordering in each case.
3. *Refusal as an outcome.* Where and why models decline to decide—by category, by explicitness, and by vendor—and what happens to applicants they decline.
4. *Representations and steering.* In open models, whether caregiving signals activate internal gender representations, how alignment changes those representations, and whether steering them changes decisions (Study 2 extension).
5. *Human comparison.* The same unbundling design with human evaluators, to compare which components humans and models penalize on identical materials—the clearest way to say whether models reproduce, transform, or exceed the human penalty.
6. *Realistic signals and deployed systems.* Weaker, naturally phrased caregiving signals; larger models; commercial screening products with their own system prompts; and other labor markets and languages, including markets in which explicit gender and age fields are standard.
7. *Group consequences with real prevalence.* Estimates of how often women and men applicants carry caregiving signals, from household surveys, to turn the disparate-impact scenario into an estimate.

8 Conclusion

Asked whether LLM résumé screeners penalize mothers, Study 1's answer is that the models we studied penalize caregivers. They penalize time out of work and visible family responsibility regardless of gender, and only when applicants are compared, while handling explicitly named gender with care that differs from vendor to vendor. We have argued that this is what protection that follows the label looks like: systems trained around a protected category treat the category carefully and its correlates freely. Because the correlates of motherhood are carried mostly by women, such systems can pass fairness checks organized around gender while reproducing the motherhood penalty in effect. Whether alignment is the cause is the question Study 2 is designed to answer.

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Items marked ◇ are to be checked against the source before submission; items marked † were added in the manuscript and are not yet in the report's verified list.

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